

AAE 50700: Principles of Dynamics
Problem Set 5 Report

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(5.1) Orbital Dynamics with J_2 and Atmospheric Drag

(a): Given

Consider a spacecraft of mass m orbiting Earth perturbed by J_2 and atmosphere drag. Let the inertial frame be $\mathcal{N} : \{\hat{e}_x, \hat{e}_y, \hat{e}_z\}$ and write its Cartesian position as $\mathbf{r} = x\hat{e}_x + y\hat{e}_y + z\hat{e}_z, r = \|\mathbf{r}\|_2$. Expressing the potential energy for Earth gravity under J_2 in the Cartesian coordinates, we have

$$V(x, y, z) = -\frac{\mu m}{r} + \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2), \quad (1)$$

where μ is Earth's gravitational parameter and $r_o = R_{\text{earth}}$ is its equatorial radius.

We use the same model as considered in PS1 to model the drag, which assumes an inertially stationary, spherically symmetric atmosphere, given by

$$\mathbf{F}_D = m\mathbf{a}_D, \quad \mathbf{a}_D = -\frac{1}{2}C_D \frac{A}{m} \rho(r) v^2 \hat{\mathbf{v}}, \quad v := \|\mathbf{v}\|_2, \quad \hat{\mathbf{v}} := \frac{\mathbf{v}}{v}, \quad (2)$$

where $\mathbf{v} = \dot{\mathbf{r}}$, C_D is the drag coefficient, A is the reference area, and

$$\rho(r) \approx \rho_0 \exp \left[-\frac{r - 6978 \text{ km}}{H} \right]. \quad (3)$$

(a.1): Find

From the Lagrangian using the Cartesian coordinates as generalized coordinates, $\mathbf{q} = [x, y, z]^T$.

Analysis/Conclusions:

First recall that the kinetic energy T of an object in motion can be expressed by half the mass times the square of the velocity:

$$T(\mathbf{v}) = \frac{m}{2} \mathbf{v}^T \mathbf{v} = \frac{mv^2}{2} \quad (4)$$

Then, recall the Lagrangian L is the kinetic energy minus the potential energy $L = T - V$ (V given by the problem definition):

$$L(\mathbf{r}, \mathbf{v}) = \frac{mv^2}{2} + \frac{\mu m}{r} - \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \quad (5)$$

(a.2): Find

Derive the generalized force for Eq. (2) associated with $\mathbf{q} = [x, y, z]^T$.

Analysis/Conclusions:

It is known that virtual work δw is translational force $\mathbf{F}$ dotted with the Jacobian of position with respect to the generalized coordinates multiplied by the displacement of generalized coordinates. This also equals generalized force $\mathbf{Q}$ dotted with the same displacement:

$$\delta w = \mathbf{F}^T \frac{\partial \mathbf{r}}{\partial \mathbf{q}} \delta \mathbf{q} = \mathbf{Q}^T \delta \mathbf{q}. \quad (6)$$

Since in this case the generalized coordinates are the Cartesian position, $\mathbf{q} = \mathbf{r}$, the Jacobian is the identity matrix, and the generalized force is simply the applied force:

$$\delta w = \mathbf{F}^T \frac{\partial \mathbf{r}}{\partial \mathbf{r}} \delta \mathbf{r} = \mathbf{F}^T I \delta \mathbf{r} = \mathbf{F}^T \delta \mathbf{r} \Rightarrow \mathbf{Q} = \mathbf{F}. \quad (7)$$

So, from Eq. (2) there is only one force, which is the force of drag $\mathbf{F}_D$. This means the generalized force from that equation associated with $\mathbf{q}$ is $\mathbf{Q} = \mathbf{F}_D$. The mass terms cancel and simplify to get generalized force:

$$\mathbf{Q} = \mathbf{F}_D = m \mathbf{a}_D = m \left(-\frac{1}{2} C_D \frac{A}{m} \rho(r) v^2 \hat{\mathbf{v}} \right) = -\frac{1}{2} C_D A \rho(r) v^2 \hat{\mathbf{v}} \quad (8)$$

(a.3): Find

Derive the Lagrange's equations and write the result as the Cartesian vector EOM

$$\ddot{\mathbf{r}} = \mathbf{a}_{2B} + \mathbf{a}_{J2} + \mathbf{a}_D \quad (9)$$

Analysis/Conclusions:

We see here that this system has three degrees of freedom with no redundancy in coordinates, so $\mathbf{q}$ is a minimal coordinate set, and the system has no constraints on movement. Since there is also a non-conservative force (drag), the general form of Lagrange's equations with non-conservative general force may be used:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial L}{\partial \mathbf{q}} = \mathbf{Q}_{nc} \quad (10)$$

And since, $\mathbf{q} = \mathbf{r}$ and $\mathbf{F}_D$ is the only non-conservative general force:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{v}}} \right) - \frac{\partial L}{\partial \mathbf{r}} = \mathbf{F}_D \quad (11)$$

Then we compute the derivatives in Eq. (11) individually.

So first to solve $\frac{\partial L}{\partial \mathbf{v}}$:

$$\begin{aligned}\frac{\partial L}{\partial \mathbf{v}} &= \frac{\partial}{\partial \mathbf{v}} \left(\frac{mv^2}{2} + \frac{\mu m}{r} - \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= \frac{m}{2} \frac{\partial v^2}{\partial \mathbf{v}} + \frac{\partial}{\partial \mathbf{v}} \left(\frac{\mu m}{r} \right) - \frac{\partial}{\partial \mathbf{v}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= \frac{m}{2} 2\mathbf{v}^T + 0 - 0 = m\mathbf{v}^T\end{aligned}\tag{12}$$

Then to solve $\frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right)$:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right) = m \frac{d\mathbf{v}^T}{dt} = m\mathbf{a}^T\tag{13}$$

Then solve $-\frac{\partial L}{\partial \mathbf{r}}$:

$$\begin{aligned}-\frac{\partial L}{\partial \mathbf{r}} &= -\frac{\partial}{\partial \mathbf{r}} \left(\frac{mv^2}{2} + \frac{\mu m}{r} - \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= -\frac{\partial}{\partial \mathbf{r}} \left(\frac{mv^2}{2} \right) - \mu m \frac{\partial r^{-1}}{\partial \mathbf{r}} + \frac{\partial}{\partial \mathbf{r}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= 0 - \mu m \frac{\partial r^{-1}}{\partial \mathbf{r}} + \frac{\partial}{\partial \mathbf{r}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right)\end{aligned}\tag{14}$$

Then for readability, we do the remaining component derivatives individually:

$$\frac{\partial r^{-1}}{\partial \mathbf{r}} = \frac{\partial r^{-1}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} = -r^{-2} \hat{\mathbf{r}}^T\tag{15}$$

And the other component:

$$\begin{aligned}
&= \frac{\partial}{\partial \mathbf{r}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(\frac{\partial}{\partial \mathbf{r}} (3r^{-5} z^2) - \frac{\partial r^{-3}}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left(\frac{\partial r^{-5}}{\partial \mathbf{r}} z^2 + r^{-5} \frac{\partial z^2}{\partial \mathbf{r}} \right) - \frac{\partial r^{-3}}{\partial \mathbf{r}} \frac{\partial r}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left(\frac{\partial r^{-5}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} z^2 + r^{-5} \frac{\partial z^2}{\partial z} \frac{\partial z}{\partial \mathbf{r}} \right) - \frac{\partial r^{-3}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left(\frac{\partial r^{-5}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} z^2 + r^{-5} \frac{\partial z^2}{\partial z} \hat{\mathbf{e}}_z^T \frac{\partial \mathbf{r}}{\partial \mathbf{r}} \right) - \frac{\partial r^{-3}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left((-5r^{-6})(\hat{\mathbf{r}}^T) z^2 + 2r^{-5} z \hat{\mathbf{e}}_z^T I \right) - (-3r^{-4} \hat{\mathbf{r}}^T) \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left((-5r^{-6})(r^{-1} \mathbf{r}^T) z^2 + 2r^{-5} z \hat{\mathbf{e}}_z^T I \right) - (-3r^{-4})(r^{-1} \mathbf{r}^T) \right) \\
&= \frac{3}{2} \mu m J_2 r_o^2 \left(-5r^{-7} z^2 \mathbf{r}^T + 2r^{-5} z \hat{\mathbf{e}}_z^T + r^{-5} \mathbf{r}^T \right) \\
&= \frac{3}{2} \mu m J_2 r_o^2 r^{-5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right) \\
&= \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right)
\end{aligned} \tag{16}$$

So then that gives:

$$\begin{aligned}
-\frac{\partial L}{\partial \mathbf{r}} &= -\mu m \left(-r^{-2} \hat{\mathbf{r}}^T \right) + \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right) \\
&= \frac{\mu m}{r^3} \mathbf{r}^T + \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right)
\end{aligned} \tag{17}$$

Which then plugging the above terms back into Eq. (11) gives:

$$m \mathbf{a}^T + \frac{\mu m}{r^3} \mathbf{r}^T + \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right) = -\frac{1}{2} C_D A \rho(r) v^2 \hat{\mathbf{v}}^T \tag{18}$$

Which then solved for acceleration $\mathbf{a} = \ddot{\mathbf{r}}$:

$$\begin{aligned}
m \mathbf{a} &= -\frac{\mu m}{r^3} \mathbf{r} - \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z \hat{\mathbf{e}}_z \right) - \frac{1}{2} C_D A \rho(r) v^2 \hat{\mathbf{v}} \\
\mathbf{a} &= -\frac{\mu m}{r^3 m} \mathbf{r} - \frac{3\mu m J_2 r_o^2}{2r^5 m} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z \hat{\mathbf{e}}_z \right) - \frac{1}{2m} C_D A \rho(r) v^2 \hat{\mathbf{v}} \\
\mathbf{a} &= -\frac{\mu}{r^3} \mathbf{r} - \frac{3\mu J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z \hat{\mathbf{e}}_z \right) - \frac{1}{2m} C_D A \rho(r) v^2 \hat{\mathbf{v}}
\end{aligned} \tag{19}$$

Which means the Cartesian vector EOM are:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} - \frac{3\mu J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z\hat{\mathbf{e}}_z \right) - \frac{1}{2m}C_D A \rho(r) v^2 \hat{\mathbf{v}} \quad (20)$$

Where:

$$\begin{aligned} \mathbf{a}_{2B} &= -\frac{\mu}{r^3}\mathbf{r} \\ \mathbf{a}_{J2} &= -\frac{3\mu J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z\hat{\mathbf{e}}_z \right) \\ \mathbf{a}_D &= -\frac{1}{2m}C_D A \rho(r) v^2 \hat{\mathbf{v}} \end{aligned} \quad (21)$$

Since these terms come from the two-body problem, J_2 perturbation, and drag acceleration components of the system respectively.

(b): Given

Now consider spherical coordinates $\mathbf{q} = [r, \phi, \theta]^T$, where r is radius, ϕ is azimuth about $\hat{\mathbf{e}}_z$, and θ is elevation measured from $\hat{\mathbf{e}}_x - \hat{\mathbf{e}}_y$ plane. The potential energy for gravity with J_2 is given by:

$$V = -\frac{m\mu}{r} \left[1 - J_2 \left(\frac{r_o}{r} \right)^2 P_2(\sin \theta) \right] \quad (22)$$

(b.1): Find

Form the Lagrangian using spherical coordinates.

Analysis/Conclusions:

The Lagrangian is once again found by kinetic minus potential energy $L = T - V$. The gravity potential including J_2 zonal harmonics was already given by Eq. (22), then the kinetic energy is once again given by:

$$T = \frac{mv^2}{2} \quad (23)$$

To express T in spherical coordinates, we write the position as $\mathbf{r} = r\hat{\mathbf{e}}_r$ in the spherical basis $\mathcal{S} : \{\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\phi, \hat{\mathbf{e}}_\theta\}$. Since this basis rotates relative to the inertial basis $\mathcal{N} : \{\hat{\mathbf{e}}_x, \hat{\mathbf{e}}_y, \hat{\mathbf{e}}_z\}$, we find $\mathbf{v}$ with the transport theorem, which requires the rotation of the spherical basis with respect to the inertial Cartesian basis, $\boldsymbol{\omega}_{\mathcal{S}/\mathcal{N}}$. We can represent this by considering $\dot{\phi}$ scaled by the $\hat{\mathbf{e}}_r$ and $\hat{\mathbf{e}}_\theta$ components of $\hat{\mathbf{e}}_z = \sin \theta \hat{\mathbf{e}}_r + \cos \theta \hat{\mathbf{e}}_\theta$, as well as $\dot{\theta}$ scaled by the negative direction of $\hat{\mathbf{e}}_\phi$:

$$\boldsymbol{\omega}_{\mathcal{S}/\mathcal{N}} = \dot{\phi} \sin \theta \hat{\mathbf{e}}_r + \dot{\phi} \cos \theta \hat{\mathbf{e}}_\theta - \dot{\theta} \hat{\mathbf{e}}_\phi \quad (24)$$

Which we then find $\mathbf{v}$ via the transport theorem as:

$$\begin{aligned} \mathbf{v} &= \frac{{}^S d}{dt} \mathbf{r} + \boldsymbol{\omega}_{\mathcal{S}/\mathcal{N}} \times \mathbf{r} \\ &= \frac{{}^S d}{dt} (r\hat{\mathbf{e}}_r) + \boldsymbol{\omega}_{\mathcal{S}/\mathcal{N}} \times (r\hat{\mathbf{e}}_r) \\ &= \frac{{}^S dr}{dt} \hat{\mathbf{e}}_r + r \frac{{}^S d\hat{\mathbf{e}}_r}{dt} + r \left(\dot{\phi} \sin \theta \hat{\mathbf{e}}_r + \dot{\phi} \cos \theta \hat{\mathbf{e}}_\theta - \dot{\theta} \hat{\mathbf{e}}_\phi \right) \times \hat{\mathbf{e}}_r \\ &= \dot{r} \hat{\mathbf{e}}_r + r(0) + r \left(\dot{\phi} \sin \theta (\mathbf{0}) + \dot{\phi} \cos \theta \hat{\mathbf{e}}_\phi - \dot{\theta} (-\hat{\mathbf{e}}_\theta) \right) \\ &= \dot{r} \hat{\mathbf{e}}_r + r \cos \theta \dot{\phi} \hat{\mathbf{e}}_\phi + r \dot{\theta} \hat{\mathbf{e}}_\theta \end{aligned} \quad (25)$$

Then we can simply use the $\mathcal{S}$ -frame representation of $\mathbf{v}$ to find an expression

for the kinetic energy T :

$$\begin{aligned}
T &= \frac{mv^2}{2} = \frac{m}{2} \mathbf{v}^T \mathbf{v} = \frac{m}{2} (v_r^2 + v_\phi^2 + v_\theta^2) \\
&= \frac{m}{2} \left(\dot{r}^2 + (r \cos \theta \dot{\phi})^2 + (r \dot{\theta})^2 \right) \\
&= \frac{m}{2} \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right)
\end{aligned} \tag{26}$$

So, the Lagrangian can be written as:

$$L = \frac{m}{2} \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right) + \frac{m\mu}{r} \left[1 - J_2 \left(\frac{r_o}{r} \right)^2 P_2(\sin \theta) \right] \tag{27}$$

However, this expression still needs to have substituted in the second degree, zeroth order, Legendre polynomial $P_2(\sin \theta)$, which can be found by the following formula:

$$P_n(x) = \sum_{m=0}^M (-1)^m \frac{(2n-2m)!}{2^n m! (n-m)! (n-2m)!} x^{n-2m} \tag{28}$$

Where n is degree and $M = n/2$.

Evaluating it becomes:

$$\begin{aligned}
P_2(\sin \theta) &= \sum_{m=0}^1 (-1)^m \frac{(2(2)-2m)!}{2^2 m! (2-m)! (2-2m)!} \sin^{2-2m} \theta \\
&= (-1)^0 \frac{(4-0)!}{2^2 (0)! (2-0)! (2-0)!} \sin^2 \theta + (-1)^1 \frac{(4-2)!}{2^2 (1)! (2-1)! (2-2)!} \sin^0 \theta \\
&= \frac{24}{4 \cdot 1 \cdot 2 \cdot 2} \sin^2 \theta - \frac{2}{4 \cdot 1 \cdot 1 \cdot 1} \\
&= \frac{3}{2} \sin^2 \theta - \frac{1}{2}
\end{aligned} \tag{29}$$

So then substituting into the found Lagrangian, we get its proper form:

$$\begin{aligned}
L &= \frac{m}{2} \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right) + \frac{m\mu}{r} \left(1 - J_2 \left(\frac{r_o}{r} \right)^2 \left(\frac{3}{2} \sin^2 \theta - \frac{1}{2} \right) \right) \\
&= \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{m\mu J_2 r_o^2}{r^3} \left(\frac{3}{2} \sin^2 \theta - \frac{1}{2} \right) \\
&= \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3}
\end{aligned} \tag{30}$$

(b.2): Find

Derive the generalized force for Eq. (2) associated with $\mathbf{q} = [r, \phi, \theta]^T$.

Analysis/Conclusions:

In part (a.2), $\mathbf{q} = \mathbf{r}$ made the Jacobian in $\mathbf{Q} = \mathbf{F}_D^T \frac{\partial \mathbf{r}}{\partial \mathbf{q}}$ the identity matrix, but now that $\mathbf{q} = [r, \phi, \theta]^T$, we must find how the position changes with respect to r , ϕ , and θ .

Let us consider the total derivative of $\mathbf{r}$ from the perspective of the chain rule:

$$\frac{d\mathbf{r}}{dt} = \frac{\partial \mathbf{r}}{\partial r} \dot{r} + \frac{\partial \mathbf{r}}{\partial \phi} \dot{\phi} + \frac{\partial \mathbf{r}}{\partial \theta} \dot{\theta} = \frac{\partial \mathbf{r}}{\partial \mathbf{q}} \dot{\mathbf{q}} \quad (31)$$

Then put it side by side with the expression found for $\mathbf{v} = \frac{d\mathbf{r}}{dt}$:

$$\mathbf{v} = \hat{\mathbf{e}}_r \dot{r} + r \cos \theta \hat{\mathbf{e}}_\phi \dot{\phi} + r \hat{\mathbf{e}}_\theta \dot{\theta} \quad (32)$$

Thus from this we see that:

$$\frac{\partial \mathbf{r}}{\partial r} = \hat{\mathbf{e}}_r, \quad \frac{\partial \mathbf{r}}{\partial \phi} = r \cos \theta \hat{\mathbf{e}}_\phi, \quad \frac{\partial \mathbf{r}}{\partial \theta} = r \hat{\mathbf{e}}_\theta \quad (33)$$

So when we compute the generalized force using $Q_j = \mathbf{F}_D \cdot \frac{\partial \mathbf{r}}{\partial q_j}$

$$\begin{aligned} Q_r &= \mathbf{F}_D \cdot \frac{\partial \mathbf{r}}{\partial r} = \left(-\frac{1}{2} C_D A \rho(r) v \mathbf{v} \right) \cdot \hat{\mathbf{e}}_r \\ &= -\frac{1}{2} C_D A \rho(r) v \left(\hat{\mathbf{e}}_r \dot{r} + r \cos \theta \hat{\mathbf{e}}_\phi \dot{\phi} + r \hat{\mathbf{e}}_\theta \dot{\theta} \right) \cdot \hat{\mathbf{e}}_r \\ &= -\frac{1}{2} C_D A \rho(r) v \dot{r} \\ Q_\phi &= \mathbf{F}_D \cdot \frac{\partial \mathbf{r}}{\partial \phi} = \left(-\frac{1}{2} C_D A \rho(r) v \mathbf{v} \right) \cdot (r \cos \theta \hat{\mathbf{e}}_\phi) \\ &= -\frac{1}{2} C_D A \rho(r) v \left(\hat{\mathbf{e}}_r \dot{r} + r \cos \theta \hat{\mathbf{e}}_\phi \dot{\phi} + r \hat{\mathbf{e}}_\theta \dot{\theta} \right) \cdot (r \cos \theta \hat{\mathbf{e}}_\phi) \\ &= -\frac{1}{2} C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi} \\ Q_\theta &= \mathbf{F}_D \cdot \frac{\partial \mathbf{r}}{\partial \theta} = \left(-\frac{1}{2} C_D A \rho(r) v \mathbf{v} \right) \cdot (r \hat{\mathbf{e}}_\theta) \\ &= -\frac{1}{2} C_D A \rho(r) v \left(\hat{\mathbf{e}}_r \dot{r} + r \cos \theta \hat{\mathbf{e}}_\phi \dot{\phi} + r \hat{\mathbf{e}}_\theta \dot{\theta} \right) \cdot (r \hat{\mathbf{e}}_\theta) \\ &= -\frac{1}{2} C_D A \rho(r) v r^2 \dot{\theta} \end{aligned} \quad (34)$$

Which makes the generalized force:

$$\begin{aligned} Q_r &= -\frac{1}{2}C_D A \rho(r) v \dot{r} \\ Q_\phi &= -\frac{1}{2}C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi} \\ Q_\theta &= -\frac{1}{2}C_D A \rho(r) v r^2 \dot{\theta} \end{aligned} \quad (35)$$

(b.3): Find

Derive the Lagrange's equations and write the result as $\ddot{r}=?$, $\ddot{\phi}=?$, and $\ddot{\theta}=?$.

Analysis/Conclusions:

Once again this system has a minimal coordinate set and a non-conservative force from drag. So, the general form of Lagrange's equations with non-conservative general force may be used:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial L}{\partial \mathbf{q}} = \mathbf{Q}_{nc} \quad (36)$$

First, we evaluate the generalized momentum terms:

$$\begin{aligned} \frac{\partial L}{\partial \dot{\phi}} &= \frac{\partial}{\partial \dot{\phi}} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\ &= 0 + \frac{mr^2 \cos^2 \theta}{2} \frac{\partial \dot{\phi}^2}{\partial \dot{\phi}} + 0 + 0 - 0 + 0 = mr^2 \dot{\phi} \cos^2 \theta \end{aligned} \quad (37)$$

And the θ momentum:

$$\begin{aligned} \frac{\partial L}{\partial \dot{\theta}} &= \frac{\partial}{\partial \dot{\theta}} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\ &= 0 + 0 + \frac{mr^2}{2} \frac{\partial \dot{\theta}^2}{\partial \dot{\theta}} + 0 - 0 + 0 = mr^2 \dot{\theta} \end{aligned} \quad (38)$$

And the r momentum:

$$\begin{aligned} \frac{\partial L}{\partial \dot{r}} &= \frac{\partial}{\partial \dot{r}} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\ &= \frac{m}{2} \frac{\partial \dot{r}^2}{\partial \dot{r}} + 0 + 0 + 0 - 0 + 0 = m\dot{r} \end{aligned} \quad (39)$$

Then we compute their time derivatives:

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) &= \frac{d}{dt} (mr^2 \dot{\phi} \cos^2 \theta) \\
&= m \left(\dot{\phi} \cos^2 \theta \frac{dr^2}{dt} + r^2 \cos^2 \theta \frac{d\dot{\phi}}{dt} + r^2 \dot{\phi} \frac{d}{dt} \cos^2 \theta \right) \\
&= m \left(\dot{\phi} \cos^2 \theta \frac{dr^2}{dr} \frac{dr}{dt} + r^2 \cos^2 \theta \frac{d\dot{\phi}}{dt} + r^2 \dot{\phi} \frac{d}{d\theta} (\cos^2 \theta) \frac{d\theta}{dt} \right) \quad (40) \\
&= m \left(2r\dot{r}\dot{\phi} \cos^2 \theta + r^2 \ddot{\phi} \cos^2 \theta - 2r^2 \dot{\phi} \dot{\theta} \cos \theta \sin \theta \right) \\
&= mr \cos \theta \left(2\dot{r}\dot{\phi} \cos \theta + r\ddot{\phi} \cos \theta - 2r\dot{\phi}\dot{\theta} \sin \theta \right)
\end{aligned}$$

And the θ time derivative:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = \frac{d}{dt} (mr^2 \dot{\theta}) = m \left(\frac{dr^2}{dr} \frac{dr}{dt} \dot{\theta} + r^2 \frac{d\dot{\theta}}{dt} \right) = 2mr\dot{r}\dot{\theta} + mr^2 \ddot{\theta} = mr (2\dot{r}\dot{\theta} + r\ddot{\theta}) \quad (41)$$

And the r time derivative:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) = m \frac{dr}{dt} = m\ddot{r} \quad (42)$$

Since the Lagrangian is not a function of ϕ :

$$\frac{\partial L}{\partial \phi} = 0 \quad (43)$$

Then compute the θ derivative:

$$\begin{aligned}
\frac{\partial L}{\partial \theta} &= \frac{\partial}{\partial \theta} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\
&= 0 + \frac{mr^2 \dot{\phi}^2}{2} \frac{\partial}{\partial \theta} (\cos^2 \theta) + 0 + 0 - \frac{3m\mu J_2 r_o^2}{2r^3} \frac{\partial}{\partial \theta} (\sin^2 \theta) + 0 \\
&= -mr^2 \dot{\phi}^2 \cos \theta \sin \theta - 3m\mu J_2 r_o^2 r^{-3} \cos \theta \sin \theta \\
&= -m \cos \theta \sin \theta \left(r^2 \dot{\phi}^2 + \frac{3\mu J_2 r_o^2}{r^3} \right) \quad (44)
\end{aligned}$$

And the r derivative:

$$\begin{aligned}
&= \frac{\partial L}{\partial r} \\
&= \frac{\partial}{\partial r} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\
&= 0 + \frac{m \cos^2 \theta \dot{\phi}^2}{2} \frac{dr^2}{dr} + \frac{m\dot{\theta}^2}{2} \frac{dr^2}{dr} + m\mu \frac{dr^{-1}}{dr} - \frac{3m\mu J_2 r_o^2}{2} \sin^2 \theta \frac{dr^{-3}}{dr} + \frac{m\mu J_2 r_o^2}{2} \frac{dr^{-3}}{dr} \\
&= mr\dot{\phi}^2 \cos^2 \theta + mr\dot{\theta}^2 - m\mu r^{-2} + \frac{9}{2}m\mu J_2 r_o^2 r^{-4} \sin^2 \theta - \frac{3}{2}m\mu J_2 r_o^2 r^{-4} \\
&= mr \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \mu r^{-3} + \frac{9}{2}\mu J_2 r_o^2 r^{-5} \sin^2 \theta - \frac{3}{2}\mu J_2 r_o^2 r^{-5} \right) \\
&= mr \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3\mu J_2 r_o^2}{2r^5} (3 \sin^2 \theta - 1) \right)
\end{aligned} \tag{45}$$

So, as per Eq. (36), we can then equate the above calculated terms to their respective non-conservative generalized drag force to get the equations of motion.

For the coordinate ϕ we set up the EoM as:

$$\begin{aligned}
&\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \frac{\partial L}{\partial \phi} = Q_{\phi,nc} \\
&mr \cos \theta \left(2\dot{r}\dot{\phi} \cos \theta + r\ddot{\phi} \cos \theta - 2r\dot{\phi}\dot{\theta} \sin \theta \right) - 0 = -\frac{1}{2}C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi}
\end{aligned} \tag{46}$$

And we then isolate the $\ddot{\phi}$ term:

$$\begin{aligned}
\ddot{\phi} m r^2 \cos^2 \theta &= -\frac{1}{2}C_D A \rho(r) v \dot{\phi} r^2 \cos^2 \theta - 2\dot{r}\dot{\phi} m r \cos^2 \theta + 2\dot{\phi}\dot{\theta} \sin \theta m r^2 \cos \theta \\
\ddot{\phi} &= -\frac{1}{2}C_D A \rho(r) v \dot{\phi} \frac{r^2 \cos^2 \theta}{m r^2 \cos^2 \theta} - 2\dot{r}\dot{\phi} \frac{m r \cos^2 \theta}{m r^2 \cos^2 \theta} + 2\dot{\phi}\dot{\theta} \frac{\sin \theta}{\cos \theta} \frac{m r^2 \cos \theta}{m r^2 \cos \theta} \\
\ddot{\phi} &= -\frac{C_D A \rho(r) v \dot{\phi}}{2m} - \frac{2\dot{r}\dot{\phi}}{r} + 2\dot{\phi}\dot{\theta} \tan \theta
\end{aligned} \tag{47}$$

For the coordinate θ we set up the EoM as:

$$\begin{aligned}
&\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = Q_{\theta,nc} \\
&mr \left(2\dot{r}\dot{\theta} + r\ddot{\theta} \right) + m \cos \theta \sin \theta \left(r^2 \dot{\phi}^2 + \frac{3\mu J_2 r_o^2}{r^3} \right) = -\frac{1}{2}C_D A \rho(r) v r^2 \dot{\theta}
\end{aligned} \tag{48}$$

And we then isolate the $\ddot{\theta}$ term:

$$\begin{aligned}
\ddot{\theta} m r^2 &= -\frac{1}{2} C_D A \rho(r) v \dot{\theta} r^2 - 2 \dot{r} \dot{\theta} m r - \dot{\phi}^2 \cos \theta \sin \theta m r^2 - 3 \mu J_2 r_o^2 \cos \theta \sin \theta \frac{m}{r^3} \\
\ddot{\theta} &= -\frac{1}{2} C_D A \rho(r) v \dot{\theta} \frac{r^2}{m r^2} - 2 \dot{r} \dot{\theta} \frac{m r}{m r^2} - \dot{\phi}^2 \cos \theta \sin \theta \frac{m r^2}{m r^2} - 3 \mu J_2 r_o^2 \cos \theta \sin \theta \frac{m}{r^3} \frac{1}{m r^2} \\
\ddot{\theta} &= -\frac{C_D A \rho(r) v \dot{\theta}}{2m} - \frac{2 \dot{r} \dot{\theta}}{r} - \dot{\phi}^2 \cos \theta \sin \theta - \frac{3 \mu J_2 r_o^2}{r^5} \cos \theta \sin \theta \\
\ddot{\theta} &= -\frac{C_D A \rho(r) v \dot{\theta}}{2m} - \frac{2 \dot{r} \dot{\theta}}{r} - \cos \theta \sin \theta \left(\dot{\phi}^2 + \frac{3 \mu J_2 r_o^2}{r^5} \right)
\end{aligned} \tag{49}$$

For the coordinate r we set up the EoM as:

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) - \frac{\partial L}{\partial r} &= Q_{r,nc} \\
m \ddot{r} - m r \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3 \mu J_2 r_o^2}{2 r^5} (3 \sin^2 \theta - 1) \right) &= -\frac{1}{2} C_D A \rho(r) v \dot{r}
\end{aligned} \tag{50}$$

And we then isolate the $\ddot{r}$ term:

$$\begin{aligned}
\ddot{r} m &= -\frac{1}{2} C_D A \rho(r) v \dot{r} + r \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3 \mu J_2 r_o^2}{2 r^5} (3 \sin^2 \theta - 1) \right) m \\
\ddot{r} &= -\frac{1}{2} C_D A \rho(r) v \dot{r} \frac{1}{m} + r \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3 \mu J_2 r_o^2}{2 r^5} (3 \sin^2 \theta - 1) \right) \frac{m}{m} \\
\ddot{r} &= -\frac{C_D A \rho(r) v \dot{r}}{2m} + r \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3 \mu J_2 r_o^2}{2 r^5} (3 \sin^2 \theta - 1) \right) \\
\ddot{r} &= -\frac{C_D A \rho(r) v \dot{r}}{2m} + r \dot{\phi}^2 \cos^2 \theta + r \dot{\theta}^2 - \frac{\mu}{r^2} + \frac{3 \mu J_2 r_o^2}{2 r^4} (3 \sin^2 \theta - 1)
\end{aligned} \tag{51}$$

Which makes the EoM:

$$\begin{aligned}
\ddot{r} &= -\frac{C_D A \rho(r) v \dot{r}}{2m} + r \dot{\phi}^2 \cos^2 \theta + r \dot{\theta}^2 - \frac{\mu}{r^2} + \frac{3 \mu J_2 r_o^2}{2 r^4} (3 \sin^2 \theta - 1) \\
\ddot{\theta} &= -\frac{C_D A \rho(r) v \dot{\theta}}{2m} - \frac{2 \dot{r} \dot{\theta}}{r} - \cos \theta \sin \theta \left(\dot{\phi}^2 + \frac{3 \mu J_2 r_o^2}{r^5} \right) \\
\ddot{\phi} &= -\frac{C_D A \rho(r) v \dot{\phi}}{2m} - \frac{2 \dot{r} \dot{\phi}}{r} + 2 \dot{\phi} \dot{\theta} \tan \theta
\end{aligned} \tag{52}$$

Where: $v = \sqrt{\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2}$

(b.4): Find

For zero drag, identify the cyclic coordinate and the associated conserved generalized momentum. Discuss its relation to the angular momentum about $\hat{e}_z$, denoted by H_z .

Analysis/Conclusions:

It was identified in (b.3) that the Lagrangian is not a function of ϕ . **This means that ϕ is the cyclic coordinate.**

If the force of drag is removed, this removes the only non-conservative force, and the Lagrange equation for ϕ now evaluates as:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \frac{\partial L}{\partial \phi} = 0 \Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = 0 \quad (53)$$

Because $\frac{\partial L}{\partial \phi} = 0$ as determined in (b.3), the generalized momentum p_ϕ is constant over time.

Likewise, the equation for this constant generalized momentum was also found in (b.3) Eq. (37):

$$p_\phi = \frac{\partial L}{\partial \dot{\phi}} = mr^2 \dot{\phi} \cos^2 \theta \quad (54)$$

To compare p_ϕ to H_z , we take the z component of $\mathbf{r} \times \mathbf{p}$, using the same relation between the spherical and Cartesian bases used to derive $\mathbf{v}$:

$$\hat{e}_z = \sin \theta \hat{e}_r + \cos \theta \hat{e}_\theta \quad (55)$$

So using this, with the previously found values (b.1) for $\mathbf{r}$ and $\mathbf{v}$ we get:

$$\begin{aligned} H_z &= \hat{e}_z \cdot (\mathbf{r} \times \mathbf{p}) = \hat{e}_z \cdot (\mathbf{r} \times m\mathbf{v}) \\ &= (\sin \theta \hat{e}_r + \cos \theta \hat{e}_\theta) \cdot (r\hat{e}_r \times (m\dot{r}\hat{e}_r + mr \cos \theta \dot{\phi} \hat{e}_\phi + mr\dot{\theta} \hat{e}_\theta)) \\ &= (\sin \theta \hat{e}_r + \cos \theta \hat{e}_\theta) \cdot (mr\dot{r}(\mathbf{0}) + mr^2 \cos \theta \dot{\phi} \hat{e}_\theta + mr^2 \dot{\theta}(-\hat{e}_\phi)) \\ &= \sin \theta(0) + \cos \theta (mr^2 \cos \theta \dot{\phi}) + 0 \\ &= mr^2 \dot{\phi} \cos^2 \theta = p_\phi \end{aligned} \quad (56)$$

So from this derivation we see that p_ϕ **is equivalent to the angular momentum about the $\hat{e}_z$ axis H_z .**

(b.5): Find

For zero drag, show that the Jacobi energy integral

$$\mathcal{J} := \dot{\mathbf{q}}^T \frac{\partial L}{\partial \dot{\mathbf{q}}} - L \quad (57)$$

is identical to the total energy $E = T + V$. Discuss if $\mathcal{J}$ is conserved under no drag, and why.

Analysis/Conclusions:

The total energy E computes just like the Lagrangian in Eq. (30), but the potential is added instead of subtracted. Likewise, the expression ends up looking quite similar:

$$\begin{aligned} E &= T + V \\ &= \frac{m}{2} \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right) - \frac{m\mu}{r} \left(1 - J_2 \left(\frac{r_o}{r} \right)^2 \left(\frac{3}{2} \sin^2 \theta - \frac{1}{2} \right) \right) \\ &= \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} - \frac{m\mu}{r} + \frac{m\mu J_2 r_o^2}{r^3} \left(\frac{3}{2} \sin^2 \theta - \frac{1}{2} \right) \\ &= \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} - \frac{m\mu}{r} + \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta - \frac{m\mu J_2 r_o^2}{2r^3} \end{aligned} \quad (58)$$

Then to find the Jacobi integral, first we compute $\dot{\mathbf{q}}^T \frac{\partial L}{\partial \dot{\mathbf{q}}}$. This is a dot product between the coordinate rates and the generalized momentum found in (b.3) in equations (37), (38), and (39):

$$\begin{aligned} \dot{\mathbf{q}}^T \frac{\partial L}{\partial \dot{\mathbf{q}}} &= \dot{r} \frac{\partial L}{\partial \dot{r}} + \dot{\theta} \frac{\partial L}{\partial \dot{\theta}} + \dot{\phi} \frac{\partial L}{\partial \dot{\phi}} \\ &= \dot{r} (m\dot{r}) + \dot{\theta} (mr^2 \dot{\theta}) + \dot{\phi} (mr^2 \dot{\phi} \cos^2 \theta) \\ &= m\dot{r}^2 + mr^2 \dot{\theta}^2 + mr^2 \cos^2 \theta \dot{\phi}^2 \end{aligned} \quad (59)$$

Then simply subtract the Lagrangian. We see below that this ends up flipping all the signs that were different between the Lagrangian and the total energy, and this ends up showing that the Jacobi integral is equal to the total energy:

$$\begin{aligned}
\mathcal{J} &= \dot{\mathbf{q}}^T \frac{\partial L}{\partial \dot{\mathbf{q}}} - L \\
&= \left(m\dot{r}^2 + mr^2\dot{\theta}^2 + mr^2 \cos^2 \theta \dot{\phi}^2 \right) \\
&\quad - \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2\dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\
&= \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2\dot{\theta}^2}{2} - \frac{m\mu}{r} + \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta - \frac{m\mu J_2 r_o^2}{2r^3} = E
\end{aligned} \tag{60}$$

In our zero drag system, **the Jacobi integral is conserved**. This is because the Jacobi integral represents generalized total energy, and for it to be conserved it must not change over time. It will change over time if: there are non-conservative forces breaking conservation of energy in the system, or there is time variance in the Lagrangian, since $\mathcal{J}$ is purely a function of L . There are no non-conservative forces, and we see from (b.3) Eq. (30) that L has no explicit dependence on time so $\mathcal{J}$ is conserved.

(b.6): Find

Under drag, derive the time derivative of H_z and $\mathcal{J}$. Discuss if they are conserved and why.

Analysis/Conclusions:

As discussed in (b.4), $H_z = p_\phi = \frac{\partial L}{\partial \dot{\phi}}$, thus the time derivative $\frac{d}{dt}(\frac{\partial L}{\partial \dot{\phi}})$ found in Eq. (40) is equivalent to the time derivative of H_z :

$$\frac{d}{dt}H_z = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = mr \cos \theta \left(2\dot{r}\dot{\phi} \cos \theta + r\ddot{\phi} \cos \theta - 2r\dot{\phi}\dot{\theta} \sin \theta \right) \quad (61)$$

Then, when substituting the derived EoM for $\ddot{\phi}$ from Eq. (52) into the time derivative of H_z , we see the equation reduce to:

$$\begin{aligned} \frac{dH_z}{dt} &= mr \cos \theta \left(2\dot{r}\dot{\phi} \cos \theta + r\ddot{\phi} \cos \theta - 2r\dot{\phi}\dot{\theta} \sin \theta \right) \\ &= mr \cos \theta \left(2\dot{r}\dot{\phi} \cos \theta + r \cos \theta \left(-\frac{C_D A \rho(r) v \dot{\phi}}{2m} - \frac{2\dot{r}\dot{\phi}}{r} + 2\dot{\phi}\dot{\theta} \tan \theta \right) - 2r\dot{\phi}\dot{\theta} \sin \theta \right) \\ &= mr \cos \theta \left(2\dot{r}\dot{\phi} \cos \theta - \frac{C_D A \rho(r) v r \dot{\phi} \cos \theta}{2m} - 2\dot{r}\dot{\phi} \cos \theta + 2r\dot{\phi}\dot{\theta} \sin \theta - 2r\dot{\phi}\dot{\theta} \sin \theta \right) \\ &= mr \cos \theta \left(-\frac{C_D A \rho(r) v r \dot{\phi} \cos \theta}{2m} \right) \\ &= -\frac{C_D A \rho(r) v}{2m} \left(mr^2 \dot{\phi} \cos^2 \theta \right) = -\frac{C_D A \rho(r) v}{2m} H_z \end{aligned} \quad (62)$$

The time rate of the Jacobi integral $\mathcal{J}$ is given by:

$$\frac{d\mathcal{J}}{dt} = \mathbf{Q}_{\text{nc}} \cdot \dot{\mathbf{q}} - \frac{\partial L}{\partial t} \quad (63)$$

The dot product portion of Eq. (63) is between non-conservative forces found

in Eq. (35) and the generalized coordinate rate, and it evaluates to:

$$\begin{aligned}
\mathbf{Q}_{\text{nc}} \cdot \dot{\mathbf{q}} &= [Q_r \quad Q_\phi \quad Q_\theta] \begin{bmatrix} \dot{r} \\ \dot{\phi} \\ \dot{\theta} \end{bmatrix} \\
&= \left[-\frac{1}{2}C_D A \rho(r) v \dot{r} \quad -\frac{1}{2}C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi} \quad -\frac{1}{2}C_D A \rho(r) v r^2 \dot{\theta} \right] \begin{bmatrix} \dot{r} \\ \dot{\phi} \\ \dot{\theta} \end{bmatrix} \\
&= -\frac{1}{2}C_D A \rho(r) v \dot{r}^2 - \frac{1}{2}C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi}^2 - \frac{1}{2}C_D A \rho(r) v r^2 \dot{\theta}^2 \\
&= -\frac{1}{2}C_D A \rho(r) v \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right) \\
&= -\frac{1}{2}C_D A \rho(r) v^3
\end{aligned} \tag{64}$$

The portion in the parenthesis ends up being equivalent to $\|\mathbf{v}\|_2^2$, so it simplifies to a fairly compact scalar function.

Looking at the Lagrangian L :

$$L = \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \tag{65}$$

As discussed in (b.5), there are no explicit time t terms, so the partial $\frac{\partial L}{\partial t} = 0$.

Which makes the time derivative of $\mathcal{J}$ equal to:

$$\frac{d\mathcal{J}}{dt} = \mathbf{Q}_{\text{nc}} \cdot \dot{\mathbf{q}} - \frac{\partial L}{\partial t} = -\frac{1}{2}C_D A \rho(r) v^3 - 0 = -\frac{1}{2}C_D A \rho(r) v^3 \tag{66}$$

So, to reiterate, the time derivatives of H_z and $\mathcal{J}$ are:

$$\frac{d}{dt} H_z = -\frac{C_D A \rho(r) v}{2m} H_z, \quad \frac{d}{dt} \mathcal{J} = -\frac{1}{2}C_D A \rho(r) v^3 \tag{67}$$

We see that H_z **and** $\mathcal{J}$ **are not conserved** because their rates of change are continuously negative scaling with air density and speed. Since $\rho(r) > 0$ and $v \geq 0$, the sign never flips, and both $\mathcal{J}$ and the magnitude of H_z always decrease under drag.

(c): Given

Let us now numerically simulate the orbital dynamics. For the physical parameters and initial condition, use the same values as defined in PS1, except for the initial semi-major axis, for which use 6628km (instead of 6978km).

Table 1: Reference orbit

Constants			Classical orbital elements at epoch			
μ_{earth}	R_{earth}	J_2	a	e	i	$\Omega_0, \omega_0, \nu_0$
$398600 \text{ km}^3/\text{s}^2$	6378.137 km	1.08263×10^{-3}	6628 km	0.01	45 deg	0 deg

Table 2: Drag model parameters

C_D	A/m	ρ_0	H
2.2	0.01 m ² /kg	$1.454 \times 10^{-13} \text{ kg/m}^3$	60 km

Define Reference Frames:

- $\mathcal{N}:\{\mathcal{O}, \hat{e}_x, \hat{e}_y, \hat{e}_z\}$: **Earth-centered inertial reference frame (ECI)**
An Earth-centered inertial reference frame with its origin at Earth center having $\hat{e}_z$ aligned with the Earth's rotational axis pointing towards the north pole.
- $\mathcal{S}:\{\mathcal{O}, \hat{e}_r, \hat{e}_\phi, \hat{e}_\theta\}$: **Spherical frame**
A frame with its origin at Earth center where $\hat{e}_r$ is aligned with the spacecraft's current position vector, $\hat{e}_\phi$ points in the direction of increasing azimuth ϕ about $\hat{e}_z$ (measured from $\hat{e}_x$), and $\hat{e}_\theta$ points in the direction of increasing elevation θ (measured from the equatorial $\hat{e}_x$ - $\hat{e}_y$ plane toward the north pole).
- $\mathcal{B}:\{\mathcal{O}, \hat{e}_r, \hat{e}_u, \hat{e}_h\}$: **Radial Transverse Normal frame (RTN)**
A body frame with its origin at Earth center where $\hat{e}_r$ is aligned with the spacecraft's current position vector, $\hat{e}_u$ is aligned with the spacecraft's local horizon in the direction of increasing argument of latitude $u = \omega + \nu$, and $\hat{e}_h$ is aligned with the orbit's specific angular momentum vector.

(c.1): Find

Express the initial conditions in the Cartesian state $(x, y, z, \dot{x}, \dot{y}, \dot{z})$ and spherical state $(r, \phi, \theta, \dot{r}, \dot{\phi}, \dot{\theta})$.

Analysis/Conclusions:

Converting from Keplerian orbital elements to Cartesian states in this context is fairly simple. The initial true anomaly is given as $\nu_0 = 0$, so the magnitudes of

spacecraft position $\mathbf{r}$ and velocity $\mathbf{v}$ can be calculated directly using the below equations, which include semi-major axis a , semi-latus rectum p , eccentricity e , and gravitational parameter μ .

$$r = \frac{p}{1 + e \cos \nu}, \quad p = a(1 - e^2), \quad v = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a}} \quad (68)$$

Which evaluate to:

$$\begin{aligned} p &= (6628 \text{ km})(1 - 0.01^2) = 6627.3372 \text{ km} \\ r &= \frac{6627.3372 \text{ km}}{1 + 0.01 \cos(0 \text{ rad})} = 6561.72 \text{ km} \\ v &= \sqrt{\frac{2(398600 \text{ km}^3/\text{s}^2)}{6561.72 \text{ km}} - \frac{398600 \text{ km}^3/\text{s}^2}{6628 \text{ km}}} \approx 7.8329 \text{ km/s} \end{aligned} \quad (69)$$

Since $\nu_0 = 0$, the spacecraft starts at periapsis, where the velocity is perpendicular to the position vector. So in the RTN frame, $\mathbf{r}$ lies entirely along $\hat{\mathbf{e}}_r$ and $\mathbf{v}$ lies entirely along $\hat{\mathbf{e}}_u$:

$$\begin{aligned} \mathbf{r} &= r\hat{\mathbf{e}}_r + 0\hat{\mathbf{e}}_u + 0\hat{\mathbf{e}}_h = 6561.72 \hat{\mathbf{e}}_r \text{ km} \\ \mathbf{v} &= 0\hat{\mathbf{e}}_r + v\hat{\mathbf{e}}_u + 0\hat{\mathbf{e}}_h \approx 7.8329 \hat{\mathbf{e}}_u \text{ km/s} \end{aligned} \quad (70)$$

Then the RTN frame $\mathbf{r}$ and $\mathbf{v}$ vectors can be converted to ECI frame using the DCM described below, which includes right ascension of the ascending node Ω , inclination i , and argument of latitude $u = \omega + \nu$. This DCM represents a 3-1-3 euler sequence that converts RTN to ECI coordinates.

$$\begin{bmatrix} \cos \Omega \cos u - \sin \Omega \cos i \sin u & -\cos \Omega \sin u - \sin \Omega \cos i \cos u & \sin \Omega \sin i \\ \sin \Omega \cos u + \cos \Omega \cos i \sin u & -\sin \Omega \sin u + \cos \Omega \cos i \cos u & -\cos \Omega \sin i \\ \sin i \sin u & \sin i \cos u & \cos i \end{bmatrix} \quad (71)$$

We see from table (1) that the initial right ascension Ω_0 , argument of periapsis ω_0 , and true anomaly ν_0 are all zero, and the initial inclination i_0 is 45 degrees. This makes the DCM evaluate to:

$$\begin{aligned} [NB] &= \begin{bmatrix} \cos 0 \cos 0 - \sin 0 \cos 45^\circ \sin 0 & -\cos 0 \sin 0 - \sin 0 \cos 45^\circ \cos 0 & \sin 0 \sin 45^\circ \\ \sin 0 \cos 0 + \cos 0 \cos 45^\circ \sin 0 & -\sin 0 \sin 0 + \cos 0 \cos 45^\circ \cos 0 & -\cos 0 \sin 45^\circ \\ \sin 45^\circ \sin 0 & \sin 45^\circ \cos 0 & \cos 45^\circ \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 45^\circ & -\sin 45^\circ \\ 0 & \sin 45^\circ & \cos 45^\circ \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \end{aligned} \quad (72)$$

Which is effectively just a 45 degree rotation about the $\hat{\mathbf{e}}_x$ axis.

Likewise we see this rotation applied to the RTN frame values of $\mathbf{r}$ and $\mathbf{v}$ in

Eq. (70):

$$\begin{aligned}
{}^{\mathcal{N}}\mathbf{r} &= [NB] {}^{\mathcal{B}}\mathbf{r} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} 6561.72 \\ 0 \\ 0 \end{bmatrix} \text{ km} = 6561.72 \hat{\mathbf{e}}_x \text{ km} \\
{}^{\mathcal{N}}\mathbf{v} &= [NB] {}^{\mathcal{B}}\mathbf{v} \approx \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 7.8329 \\ 0 \end{bmatrix} \text{ km/s} \approx 5.5387 \hat{\mathbf{e}}_y + 5.5387 \hat{\mathbf{e}}_z \text{ km/s}
\end{aligned} \tag{73}$$

which gives the Cartesian state initial conditions.

Then to convert to spherical coordinates, we use the following relationships between Cartesian and spherical coordinates, where r is the norm of $\mathbf{r}$ and ρ is the length of the projection of $\mathbf{r}$ onto the $\hat{\mathbf{e}}_x$ - $\hat{\mathbf{e}}_y$ plane:

$$\begin{aligned}
\rho &= r \cos \theta, & x &= \rho \cos \phi = r \cos \theta \cos \phi, \\
z &= r \sin \theta, & y &= \rho \sin \phi = r \cos \theta \sin \phi, \\
r &= \sqrt{x^2 + y^2 + z^2} = \sqrt{\rho^2 + z^2}
\end{aligned} \tag{74}$$

In Eq. (74), the conversion for the r coordinate is already found (which is just the norm of $\mathbf{r}$). Then we can derive expressions for ϕ and θ by dividing y by x and z by ρ :

$$\begin{aligned}
\frac{y}{x} &= \frac{\rho \sin \phi}{\rho \cos \phi} = \frac{\sin \phi}{\cos \phi} = \tan \phi \quad \Rightarrow \quad \phi = \tan^{-1} \left(\frac{y}{x} \right) \\
\frac{z}{\rho} &= \frac{r \sin \theta}{r \cos \theta} = \frac{\sin \theta}{\cos \theta} = \tan \theta \quad \Rightarrow \quad \theta = \tan^{-1} \left(\frac{z}{\rho} \right) = \tan^{-1} \left(\frac{z}{\sqrt{x^2 + y^2}} \right)
\end{aligned} \tag{75}$$

And by utilizing the atan2 algorithm in Matlab, these derived expressions can be used to compute these angles while resolving the quadrant ambiguity.

Finding the derivative components $\dot{r}$, $\dot{\phi}$, and $\dot{\theta}$ then requires a derivative of the

expressions used to convert from Cartesian to spherical:

$$\text{Let: } u = x^2 + y^2 + z^2$$

$$\begin{aligned}\dot{r} &= \frac{d}{dt} r = \frac{d}{du} (u^{1/2}) \frac{du}{dt} \\ &= \frac{1}{2} u^{-1/2} \frac{d}{dt} (x^2 + y^2 + z^2) \\ &= \frac{1}{2} u^{-1/2} \left(\frac{d(x^2)}{dx} \frac{dx}{dt} + \frac{d(y^2)}{dy} \frac{dy}{dt} + \frac{d(z^2)}{dz} \frac{dz}{dt} \right) \\ &= \frac{1}{2} u^{-1/2} (2x\dot{x} + 2y\dot{y} + 2z\dot{z}) \\ &= \frac{x\dot{x} + y\dot{y} + z\dot{z}}{\sqrt{x^2 + y^2 + z^2}} = \frac{x\dot{x} + y\dot{y} + z\dot{z}}{r}\end{aligned}\tag{76}$$

And the derivative to get $\dot{\phi}$:

$$\text{Let: } u = \frac{y}{x}$$

$$\begin{aligned}\dot{\phi} &= \frac{d}{dt} \phi = \frac{d}{du} (\tan^{-1} u) \frac{du}{dt} = \frac{1}{1+u^2} \frac{d}{dt} \left(\frac{y}{x} \right) = \frac{1}{1+u^2} \left(\frac{x \frac{dy}{dt} - y \frac{dx}{dt}}{x^2} \right) \\ &= \frac{1}{1 + \left(\frac{y}{x} \right)^2} \left(\frac{x\dot{y} - y\dot{x}}{x^2} \right) = \frac{x^2}{x^2 + y^2} \left(\frac{x\dot{y} - y\dot{x}}{x^2} \right) = \frac{x\dot{y} - y\dot{x}}{x^2 + y^2}\end{aligned}\tag{77}$$

And the derivative to get $\dot{\theta}$:

$$\text{Let: } u = \frac{z}{\rho}, \quad \rho = \sqrt{x^2 + y^2}, \quad w = x^2 + y^2$$

$$\begin{aligned}\dot{\rho} &= \frac{d}{dt} \rho = \frac{d}{dw} (w^{1/2}) \frac{dw}{dt} = \frac{1}{2} w^{-1/2} \left(\frac{d(x^2)}{dx} \frac{dx}{dt} + \frac{d(y^2)}{dy} \frac{dy}{dt} \right) \\ &= \frac{1}{2} w^{-1/2} (2x\dot{x} + 2y\dot{y}) = \frac{x\dot{x} + y\dot{y}}{\sqrt{x^2 + y^2}} = \frac{x\dot{x} + y\dot{y}}{\rho} \\ \dot{\theta} &= \frac{d}{dt} \theta = \frac{d}{du} (\tan^{-1} u) \frac{du}{dt} = \frac{1}{1+u^2} \frac{d}{dt} \left(\frac{z}{\rho} \right) = \frac{1}{1+u^2} \left(\frac{\rho \frac{dz}{dt} - z \frac{d\rho}{dt}}{\rho^2} \right) \\ &= \frac{1}{1 + \left(\frac{z}{\rho} \right)^2} \left(\frac{\rho\dot{z} - z\dot{\rho}}{\rho^2} \right) = \frac{\rho^2}{\rho^2 + z^2} \left(\frac{\rho\dot{z} - z\dot{\rho}}{\rho^2} \right) = \frac{\rho\dot{z} - z\dot{\rho}}{\rho^2 + z^2} = \frac{\rho\dot{z} - z\dot{\rho}}{r^2} \\ &= \frac{\rho\dot{z} - z \left(\frac{x\dot{x} + y\dot{y}}{\rho} \right)}{r^2} = \frac{\rho^2\dot{z} - z(x\dot{x} + y\dot{y})}{\rho r^2}\end{aligned}\tag{78}$$

So, we then find spherical states r , ϕ , θ , $\dot{r}$, $\dot{\phi}$, and $\dot{\theta}$:

$$\begin{aligned}
r &= \sqrt{x^2 + y^2 + z^2} = \sqrt{(6561.72)^2 + (0)^2 + (0)^2} = 6561.72 \text{ km} \\
\phi &= \text{atan2}(y, x) = \text{atan2}(0, 6561.72) = 0^\circ \\
\theta &= \text{atan2}\left(z, \sqrt{x^2 + y^2}\right) = \text{atan2}\left(0, \sqrt{(6561.72)^2 + (0)^2}\right) = 0^\circ \\
\dot{r} &= \frac{x\dot{x} + y\dot{y} + z\dot{z}}{r} = \frac{(6561.72)(0) + (0)(5.5387) + (0)(5.5387)}{6561.72} = 0 \text{ km/s} \\
\dot{\phi} &= \frac{x\dot{y} - y\dot{x}}{x^2 + y^2} = \frac{(6561.72)(5.5387) - (0)(0)}{(6561.72)^2 + (0)^2} \left(\frac{180^\circ}{\pi}\right) \approx 0.048363 \text{ deg/s} \\
\rho &= \sqrt{x^2 + y^2} = \sqrt{(6561.72)^2 + (0)^2} = 6561.72 \text{ km} \\
\dot{\theta} &= \frac{\rho^2\dot{z} - z(x\dot{x} + y\dot{y})}{\rho r^2} \\
&= \frac{(6561.72)^2(5.5387) - (0)((6561.72)(0) + (0)(5.5387))}{(6561.72)(6561.72)^2} \left(\frac{180^\circ}{\pi}\right) \approx 0.048363 \text{ deg/s}
\end{aligned} \tag{79}$$

So to reiterate, the found initial Cartesian and spherical states are:

$$\begin{bmatrix} x \\ y \\ z \\ \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} \approx \begin{bmatrix} 6561.72 \text{ km} \\ 0 \text{ km} \\ 0 \text{ km} \\ 0 \text{ km/s} \\ 5.5387 \text{ km/s} \\ 5.5387 \text{ km/s} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} r \\ \phi \\ \theta \\ \dot{r} \\ \dot{\phi} \\ \dot{\theta} \end{bmatrix} \approx \begin{bmatrix} 6561.72 \text{ km} \\ 0^\circ \\ 0^\circ \\ 0 \text{ km/s} \\ 0.048363 \text{ deg/s} \\ 0.048363 \text{ deg/s} \end{bmatrix} \tag{80}$$

(c.2): Find

Propagate the orbit for 5 days using the EOM derived for spherical coordinates, with the following two models:

- J_2 gravity only
- J_2 gravity with drag.

Plot each inertial trajectory in Cartesian coordinates separately. Also, plot time history of the difference between the two models.

Analysis/Conclusions:

The models are propagated for 5 days, which converts to seconds as:

$$t_f = 5 \text{ days} \left(\frac{24 \text{ hr}}{1 \text{ day}}\right) \left(\frac{60 \text{ min}}{1 \text{ hr}}\right) \left(\frac{60 \text{ s}}{1 \text{ min}}\right) = 432,000 \text{ s} \tag{81}$$

Then, the EoM in Eq. (52) must be put in a form compatible with the numerical integrator. This works simply if we integrate an orbital spherical coordinate state of $\mathbf{x} = [r, \phi, \theta, \dot{r}, \dot{\phi}, \dot{\theta}]^T$, whose time derivative is:

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{r} \\ \dot{\phi} \\ \dot{\theta} \\ \ddot{r} \\ \ddot{\phi} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{r} \\ \dot{\phi} \\ \dot{\theta} \\ -\frac{C_D A \rho(r) v \dot{r}}{2m} + r \dot{\phi}^2 \cos^2 \theta + r \dot{\theta}^2 - \frac{\mu}{r^2} + \frac{3\mu J_2 r_o^2}{2r^4} (3 \sin^2 \theta - 1) \\ -\frac{C_D A \rho(r) v \dot{\phi}}{2m} - \frac{2\dot{r}\dot{\phi}}{r} + 2\dot{\phi}\dot{\theta} \tan \theta \\ -\frac{C_D A \rho(r) v \dot{\theta}}{2m} - \frac{2\dot{r}\dot{\theta}}{r} - \cos \theta \sin \theta \left(\dot{\phi}^2 + \frac{3\mu J_2 r_o^2}{r^5} \right) \end{bmatrix} \quad (82)$$

Where: $v = \sqrt{\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2}$

Eq. (82) works as is for the model that includes drag force. For the model that does not include drag, we can simply set the drag coefficient to zero which gives the equations:

$$\begin{aligned} \dot{\mathbf{x}} &= \begin{bmatrix} \dot{r} \\ \dot{\phi} \\ \dot{\theta} \\ -\frac{(0)A\rho(r)v\dot{r}}{2m} + r\dot{\phi}^2 \cos^2 \theta + r\dot{\theta}^2 - \frac{\mu}{r^2} + \frac{3\mu J_2 r_o^2}{2r^4} (3 \sin^2 \theta - 1) \\ -\frac{(0)A\rho(r)v\dot{\phi}}{2m} - \frac{2\dot{r}\dot{\phi}}{r} + 2\dot{\phi}\dot{\theta} \tan \theta \\ -\frac{(0)A\rho(r)v\dot{\theta}}{2m} - \frac{2\dot{r}\dot{\theta}}{r} - \cos \theta \sin \theta \left(\dot{\phi}^2 + \frac{3\mu J_2 r_o^2}{r^5} \right) \end{bmatrix} \\ &= \begin{bmatrix} \dot{r} \\ \dot{\phi} \\ \dot{\theta} \\ r\dot{\phi}^2 \cos^2 \theta + r\dot{\theta}^2 - \frac{\mu}{r^2} + \frac{3\mu J_2 r_o^2}{2r^4} (3 \sin^2 \theta - 1) \\ -\frac{2\dot{r}\dot{\phi}}{r} + 2\dot{\phi}\dot{\theta} \tan \theta \\ -\frac{2\dot{r}\dot{\theta}}{r} - \cos \theta \sin \theta \left(\dot{\phi}^2 + \frac{3\mu J_2 r_o^2}{r^5} \right) \end{bmatrix} \end{aligned} \quad (83)$$

Which is effectively just the EoM with the drag term removed.

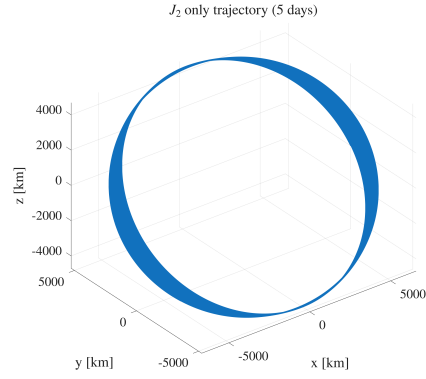
Then the only other things we need to do are convert the units in the A/m and ρ_0 terms in table (2) to use km-based units and include an expression to convert to Cartesian coordinates for plotting. The unit conversions result in:

$$\begin{aligned} \frac{A}{m} &= 0.01 \text{ m}^2/\text{kg} \left(\frac{1 \text{ km}}{1000 \text{ m}} \right)^2 = 1 \times 10^{-8} \text{ km}^2/\text{kg} \\ \rho_0 &= 1.454 \times 10^{-13} \text{ kg/m}^3 \left(\frac{1000 \text{ m}}{1 \text{ km}} \right)^3 = 1.454 \times 10^{-4} \text{ kg/km}^3 \end{aligned} \quad (84)$$

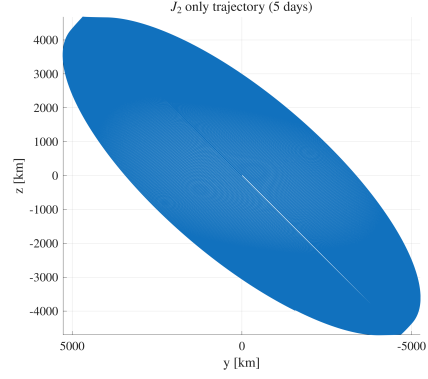
And a standard expression to map spherical coordinates to Cartesian coordinates is:

$$\mathbf{r} = r \cos \theta \cos \phi \hat{\mathbf{e}}_x + r \cos \theta \sin \phi \hat{\mathbf{e}}_y + r \sin \theta \hat{\mathbf{e}}_z \quad (85)$$

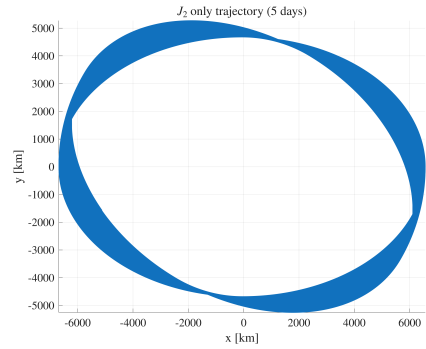
So, using the methods described above, the following plots show the J_2 only model propagated for 5 days:



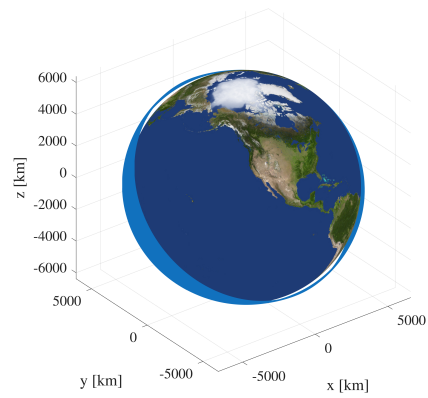
(a) 3D view of the trajectory.



(b) Edge-on view of the trajectory.



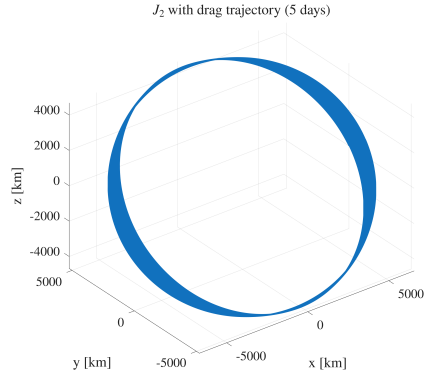
(c) Side-on view of the trajectory.



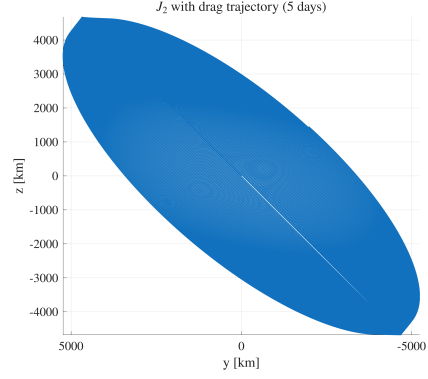
(d) Trajectory with Earth shown.

Figure 1: J_2 only trajectory over 5 days, shown from multiple viewing angles.

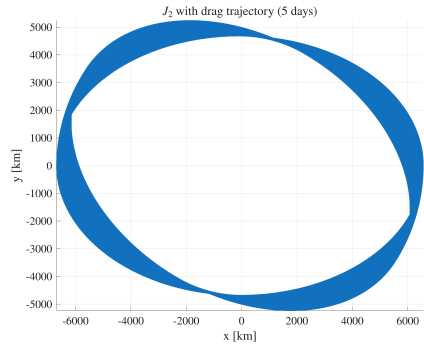
And the following plots show the J_2 with drag model propagated for 5 days:



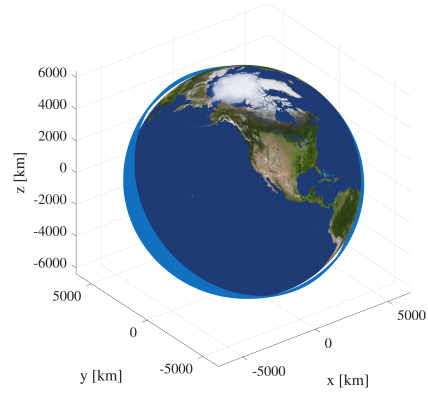
(a) 3D view of the trajectory.



(b) Edge-on view of the trajectory.



(c) Side-on view of the trajectory.



(d) Trajectory with Earth shown.

Figure 2: J_2 with drag trajectory over 5 days, shown from multiple viewing angles.

Then, to plot the time history of the position difference between the two models, we simply find the difference between the x , y , and z positions of the models over time, $\Delta \mathbf{r}(t) = \mathbf{r}_{\text{drag}}(t) - \mathbf{r}_{J_2}(t)$:

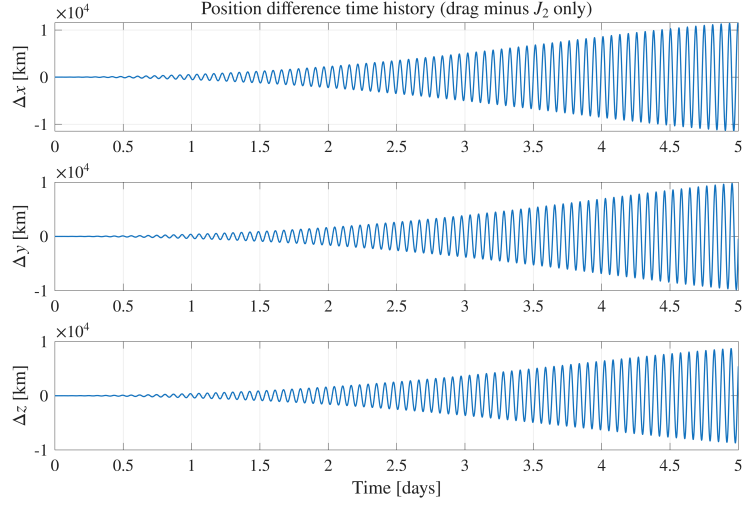


Figure 3: Time history of the position difference between the J_2 with drag and J_2 only models over 5 days.

(c.3): Find

For both models, plot the normalized changes in H_z , $\mathcal{J}$:

$$\frac{\mathcal{J}(t) - \mathcal{J}(0)}{|\mathcal{J}(0)|}, \quad \frac{H_z(t) - H_z(0)}{|H_z(0)|} \quad (86)$$

Discuss why the J_2 -only result makes sense using the symmetry of the potential. For the drag result, compare the signs and qualitative behavior against dH_z/dt and $d\mathcal{J}/dt$ derived above.

Analysis/Conclusions:

From part (b.5) in Eq. (60), the Jacobi integral for both models was found:

$$\mathcal{J} = \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} - \frac{m\mu}{r} + \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta - \frac{m\mu J_2 r_o^2}{2r^3} \quad (87)$$

And in part (b.4) in Eq. (56), the function for H_z for both models was found:

$$H_z = mr^2 \dot{\phi} \cos^2 \theta \quad (88)$$

Since the spacecraft mass m is not specified, we instead evaluate both quantities per unit mass:

$$\frac{\mathcal{J}}{m} = \frac{\dot{r}^2}{2} + \frac{r^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{r^2 \dot{\theta}^2}{2} - \frac{\mu}{r} + \frac{3\mu J_2 r_o^2}{2r^3} \sin^2 \theta - \frac{\mu J_2 r_o^2}{2r^3} \quad (89)$$

$$\frac{H_z}{m} = r^2 \dot{\phi} \cos^2 \theta$$

Utilizing the initial conditions found in (c.1) Eq. (80) and the values defined in table (1), the values of $\mathcal{J}(t)$ and $H_z(t)$ are computed over the simulation time of both models using the expressions above, starting from their initial values $\mathcal{J}(0)$ and $H_z(0)$. As requested by the problem, the differences $\mathcal{J}(t) - \mathcal{J}(0)$ and $H_z(t) - H_z(0)$ are plotted over time, normalized by the absolute value of their initial values.

The normalized changes for the J_2 only model are:

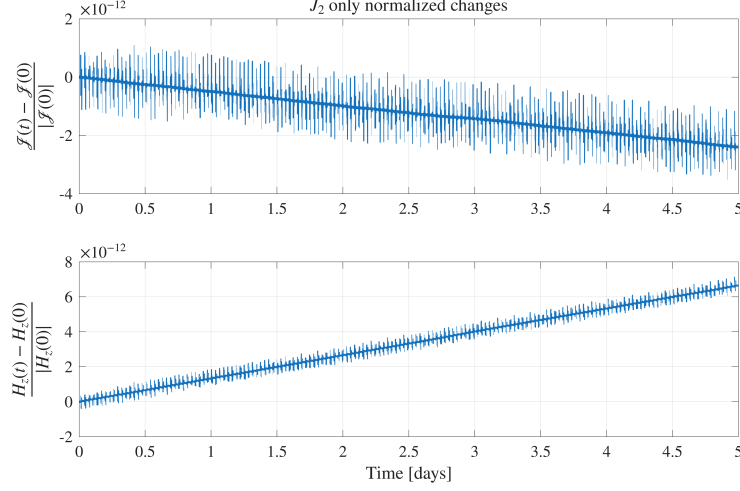


Figure 4: Normalized changes in $\mathcal{J}$ and H_z for the J_2 only model over 5 days.

We see in the results above, that the values of $\mathcal{J}$ and H_z (when normalized to their initial values) do not vary further than the range of 10^{-12} from their initial values, which is in the range of numeric drift. This can be interpreted to mean that in the no-drag model, $\mathcal{J}$ and H_z do not change over time.

The J_2 -only result makes sense here because of the symmetries of the potential.

First, for the angular momentum about $\hat{e}_z$ H_z , the J_2 potential seen in Eq. (22) depends only on r and θ , so it is symmetric about the $\hat{e}_z$ axis. This means ϕ is a cyclic coordinate, $\frac{\partial L}{\partial \phi} = 0$, and with no drag the Lagrange equation for ϕ reduces to $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = 0$. As established in (b.4), the generalized momentum $\frac{\partial L}{\partial \dot{\phi}}$ is equal to H_z , so H_z cannot change over time in this model.

Second, for the Jacobi integral, the J_2 potential Eq. (22) is also symmetric in time, since it has no explicit dependence on t , so $\frac{\partial L}{\partial t} = 0$. With no non-conservative forces, Eq. (63) then becomes $\frac{d\mathcal{J}}{dt} = 0$, so $\mathcal{J}$ cannot change over time either. Both results are consistent with the plots above, where the only variation is numerical drift.

And the normalized changes for the J_2 with drag model are:

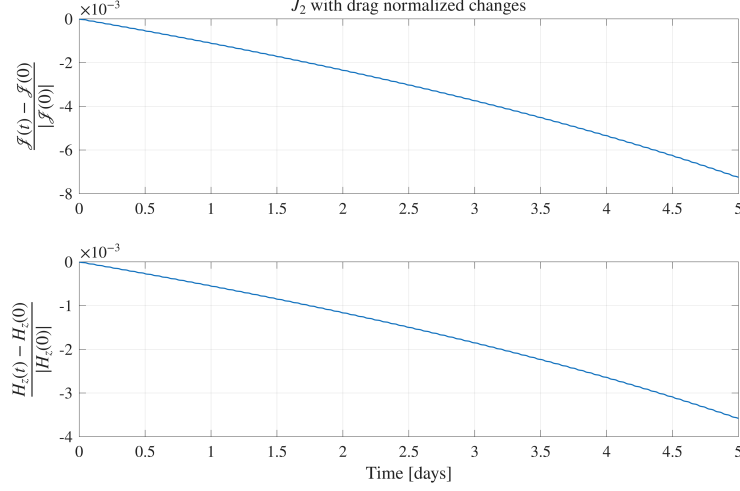


Figure 5: Normalized changes in $\mathcal{J}$ and H_z for the J_2 with drag model over 5 days.

Recall from part (b.6) the equations for dH_z/dt and $d\mathcal{J}/dt$ were derived to be:

$$\frac{d}{dt}H_z = -\frac{C_D A \rho(r) v}{2m} H_z, \quad \frac{d}{dt}\mathcal{J} = -\frac{1}{2} C_D A \rho(r) v^3 \quad (90)$$

In the analysis in part (b.6), it was identified that the quantities used in these expressions would never cause the sign to flip, meaning we should expect to see H_z and $\mathcal{J}$ to continuously decrease due to the negative sign in both time derivative expressions. This is consistent with the figure above which shows the two quantities continuously deviating in the negative direction from their initial values with a smooth curve and at an order of magnitude that would rule out numeric drift.

We also see both curves steepen over time, which is consistent with the time derivative expressions. If angular momentum and generalized total energy (aka Jacobi integral) are decreasing, then we should expect the orbit to decay, lowering the average altitude of the spacecraft along its orbit. Since the rates of H_z and $\mathcal{J}$ both scale with $\rho(r)$, which grows exponentially as altitude decreases, and with v , which increases as the orbit decays, the decay of the orbit quickens the rate at which H_z and $\mathcal{J}$ decrease. For H_z , this growth in $\rho(r)$ appears to outpace the slowing effect of its rate being proportional to H_z itself.

(c.4): Find

Augment the state vector by the work state ($\mathbf{Q}_D$: drag-derived generalized force)

$$\dot{W}_D = \mathbf{Q}_D^T \dot{\mathbf{q}}, \quad W_D(0) = 0 \quad (91)$$

and propagate the orbit under drag again. Plot the normalized work-balance residual

$$\frac{\mathcal{J}(t) - \mathcal{J}(0) - W_D(t)}{|\mathcal{J}(0)|} \quad (92)$$

Discuss whether it is consistent with your integrator tolerance.

Analysis/Conclusions:

Recall from part (b.2) the generalized drag force $\mathbf{Q}_D$ was found to be:

$$\begin{aligned} Q_r &= -\frac{1}{2} C_D A \rho(r) v \dot{r} \\ Q_\phi &= -\frac{1}{2} C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi} \\ Q_\theta &= -\frac{1}{2} C_D A \rho(r) v r^2 \dot{\theta} \end{aligned} \quad (93)$$

When performing the dot product with the generalized coordinate derivative $\dot{\mathbf{q}}$ to get the work state derivative $\dot{W}_D$, we get:

$$\begin{aligned} \dot{W}_D &= \mathbf{Q}_D^T \dot{\mathbf{q}} = [Q_r \quad Q_\phi \quad Q_\theta] \begin{bmatrix} \dot{r} \\ \dot{\phi} \\ \dot{\theta} \end{bmatrix} \\ &= -\frac{1}{2} C_D A \rho(r) v \dot{r}^2 - \frac{1}{2} C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi}^2 - \frac{1}{2} C_D A \rho(r) v r^2 \dot{\theta}^2 \\ &= -\frac{1}{2} C_D A \rho(r) v \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right) \\ &= -\frac{1}{2} C_D A \rho(r) v^3 \end{aligned} \quad (94)$$

Which is the same expression derived for $\frac{d}{dt} \mathcal{J}$, which means the residual in Eq. (92) should stay near zero within the expected range from the integration tolerance. Note that since A and m are not individually specified, only their ratio A/m , the work state is evaluated per unit mass, the same as $\mathcal{J}$ and H_z were in (c.3).

So, after running the simulation with the augmented state, the normalized work-balance residual for the J_2 with drag model is:

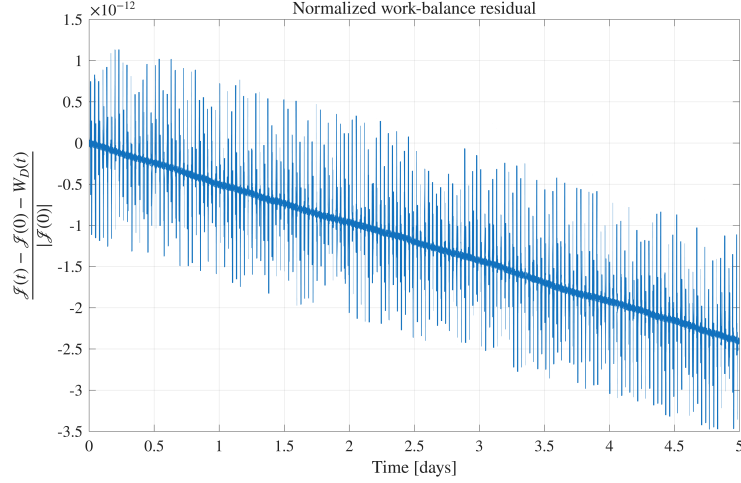


Figure 6: Normalized work-balance residual for the J_2 with drag model over 5 days.

The above results **are consistent with my integrator tolerance** of 10^{-12} because we see in the plot above the residual stays within the range of about -3.5×10^{-12} and 1.25×10^{-12} , which means the residual effectively did not vary between steps within tolerance. Since $\dot{W}_D$ is equivalent to $\frac{d\mathcal{J}}{dt}$, integrating it analytically should reproduce $\mathcal{J}(t) - \mathcal{J}(0)$ exactly, so the residual would be zero if formed from analytic integration. The small nonzero residual is therefore purely numerical integration error and is within tolerance.